

Syllabus

1st semester 2026

Course Title	General Physics 1	Course Code	PE3101		Credit	4	Classroom	Hyung3302 Hyung3305
		Type	Natural Science / Elective		Hours	4		
Teacher	Name		Class no.	Office	Office Hour*	Contact	E-mail	
	Lee, Seungkoog		1				sk890528@gmail.com	
	Sim, Jae-hoon		EC-1	Chang5502	Wed 3-6		jaehoon.sim@ksa.kaist.ac.kr	
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1. Course Description

- Newtonian Classical Mechanics: Mechanics is made up of kinematics and dynamics. Kinematics treats the description of the motion. Dynamics cares about the cause of motion, the force. The main concepts are inertial observer, point particle, motion, position, displacement, velocity, acceleration, mass, force, work, kinetic energy, potential energy, many particles system, linear momentum, torque, angular momentum, equilibrium, gravity, and fluid.
- Oscillations and Waves: We study the oscillation and wave motion of the bulk material in the viewpoint of Newtonian mechanics. The main concepts are simple harmonic motion, pendulum, superposition, diffraction and interference.
- Thermodynamics: We study the relationships between the macroscopic quantities of the many particles system. The main concepts are heat, temperature, pressure, volume and 1st law of thermodynamics.
- Kinetic Theory of Gases: Macroscopic behavior of ideal gas is analyzed in terms of microscopic dynamics.
- Statistical Mechanics: We study the macroscopic thermodynamic process by using the concept of probability. The main concepts are entropy, 2nd law of thermodynamics and Maxwell-Boltzmann distribution.

2. Core Target Competencies

Subject Core Competencies			
Critical Thinking Skills	Analytical and Creative Thinking	Complex Problem-Solving Skills	Creativity
Fundamental concepts in several subjects in physics will be presented, including Newtonian dynamics, thermodynamics, and statistical physics, are introduced. Following the homework and the discussion sessions in this lecture, students will be encouraged to develop their competencies listed above. The ways of thinking, trained in this lecture, will provide important background for future coursework.			

3. Textbooks and References

A. Textbook: University Physics with Modern Physics, 2nd Ed. (Bauer, Westfall, McGraw-Hill)

B. References

- University Physics with Modern Physics, 3rd Ed. (Bauer, Westfall, McGraw-Hill)
- The Feynman Lectures on Physics I, II, III (R. Feynman et al., Addison-Wesley)
- Berkeley Physics Course I (McGraw-Hill)

4. Class Method

- A. Self-learning while doing homework, assessment by short quizzes.
- B. In-class presentations, giving students a variety of possible ways which offer an approach to the subject different from the textbook, and in addition to the textbook.
- C. Wide use in the class learning a physical demonstrations and computer simulations of studying processes.

5. Evaluation

A. Evaluation Criteria

Midterm	Final	Quiz	Homework	Attendance and Attitude	Total
30	30	20	10	10	100

B. Evaluation Method

- (1) The midterm and final exams are given through written tests.
- (2) Quizzes will be given several times during the semester.
- (3) Homework: Penalty will be given to the overdue homework.
- (4) Attendance: -0.3 point per late & -1.0 point per absence without notice.

6. Criteria for Awarding Credits

- ☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- **Attendance for at least three-quarters of the total class hours is required to earn credit.**
- **For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.**
- **Calculation of the basis score for awarding credit to absentees: Projecting a student's performance by applying the Z-score from an attended exam to the distribution of a missed exam.**
- **If a basis score cannot be calculated, the grading method: Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.**

7. Weekly Plans

Week	Topic and Contents	Homework
Week 1 (2.23~2.27)	Ch.1 Overview Ch.2 Motion in a Straight Line	Ch1: 64, 67, 71, 80, 89 Ch2: 30, 44, 77, 80, 94
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 2 (3.01~3.06)	Ch.3 Motion in Two and Three Dimensions Ch.4 Force	Ch3: 39, 51, 60, 67, 98 Ch4: 35, 40, 51, 67, 92
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 3 (3.09~3.13)	Ch.5 Kinetic Energy, Work, and Power Ch.6 Potential Energy and Energy Conservation	Ch5: 31, 41, 52, 54, 72 Ch6: 36, 51, 62, 68, 90
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 4 (3.16~3.20)	Ch.7 Momentum and Collisions Ch.8 Systems of Particles and Extended Objects	Ch7: 33, 45, 54, 99, 108 Ch8: 32, 33, 45, 53, 75
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 5 (3.23~3.27)	Ch.9 Circular Motion Ch.10 Rotation	Ch9: 38, 47, 59, 64, 80 Ch10: 44, 54, 62, 75, 83
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 6 (3.30~4.03)	Ch.11 Static Equilibrium	Ch11: 35, 47, 63, 67, 74
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 7 (4.06~4.10)	Ch.12 Gravitation	Ch12: 39, 47, 59, 73, 80
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 8 (4.13~4.17)	Midterm Exams (4.14~4.17)	

Week	Topic and Contents	Homework
Week 9 (4.20~4.24)	Ch.13 Solids and Fluids	Ch13: 32, 39, 47, 55, 58, 72, 80
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 10 (4.27~5.01)	Ch.14 Oscillations	Ch14: 29, 37, 47, 49, 58, 70, 76
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 11 (5.04~5.08)	Ch.15 Waves	(SAF: 5.08~5.09) Ch15: 27, 33, 39, 43, 52, 70
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 12 (5.11~5.15)	Ch.16 Sound	Ch16: 33, 40, 49, 63, 69
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 13 (5.18~5.22)	Ch.17 Temperature Ch.18 Heat and the First Law of Thermodynamics	Ch17: 35, 43, 55, 61, 74 Ch18: 39, 47, 57, 66, 76
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 14 (5.25~5.29)	Ch.19 Ideal Gases	Ch19: 33, 41, 61, 71, 82
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 15 (6.01~6.05)	Ch.20 The Second Law of Thermodynamics	Ch20: 31, 43, 51, 58, 70
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 16 (6.08~6.12)	Make-up class week 3.01(Mon) -> 6.08(Mon) / 5.05(Tue) -> 6.09(Tue) / 5.25(Mon) -> 6.10(Wed) / 6.03(Wed) -> 6.11(Thu)	
Week 17 (6.15~6.19)	Final Exams	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression/ Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test	Ability to explain	Ability to collaborate on		Ability to use 21 st century core	Ability to develop a sense of	Ability to make rational and	Self-efficacy

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression/ Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
solutions to problems by trial and error	information and persuasively engage others in an engaging way	tasks, manage groups, and delegate responsibilities		competencies such as communication skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	independence to seek new roles, ideas, or strategies	ethical decisions on complex issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematic efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways